Reproducibility Report: Coppock (2021)

Visualize as You Randomize: Design-Based Statistical Graphs for Randomized Experiments

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This repository holds the actively maintained replication code for Coppock (2021), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Chapter	10.1017/9781108777919.022
Replication archive	10.7910/DVN/VE6VSR
Pre-analysis plan	none: the chapter analyses simulated data

The data are not redistributed here. The deposit is 280 KB across 17 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the Dataverse file identifiers, the UNF of each ingested data file, the deposit's own directory layout, and two checksums per file: the MD5 of the bytes Dataverse serves for `?format=original`, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all seventeen. They do not always, so the script verifies against the served bytes and reports any disagreement. Either way the exact bytes are pinned in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every claim the chapter makes about its figures to the code that produces it, and is itself built by a script rather than typed; `ground_truth/published_claims.csv` is the extraction, a hand-reviewed inventory of every number and counted quantity in the chapter. `maintained/in_text_claims.R` recomputes each of those the pipeline can reach and prints it beside the sentence that states it. `coppock_2021_errata.pdf` lists the sentences and labels in the chapter that the deposited data do not support. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2021.Rproj`, and run:

```r
source("run_all.R")
```

That fetches the deposit, verifies its seventeen checksums, produces every figure into `maintained/output/`, rebuilds the ground truth table from those outputs, runs the coverage gate over the claims file, and prints the claim-by-claim audit trail. It takes about ten seconds. Required packages: tidyverse, estimatr, broom, margins, DeclareDesign, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. Individual scripts can be run on their own in any order, with two exceptions: `text_archive_agreement.R` reads the figure scripts' output and `build_ground_truth.R` reads everything, so both come last.

A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV and PNG output should come back byte-identical. The fourteen PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Not on its own. Six of the archive's eight R scripts fail in a clean R session, and the two that survive do so by accident: `blocked.R` and `interaction.R` happen to load `estimatr`, which re-exports the `tidy()` generic the other scripts call without loading anything that provides it. Four scripts stop at `could not find function "tidy"` and one at `could not find function "lm_lin"`, all for the same reason, which is that every script opens with `library(tidyverse)` and nothing else. Adding `library(broom)` and `library(estimatr)` makes all seven analysis scripts run to completion, so the failure is a missing declaration rather than a missing package. This is what a script that has only ever been run in a warm session looks like from outside: it works for its author and stops for everybody else.

The eighth script is the data generator, and it fails for three separate reasons. `declare_assignment()` rejects the pre-1.0 DeclareDesign syntax, `draw_likert()` no longer supplies default cut points, and `declare_sampling()` rejects its own pre-1.0 syntax at the last design in the file. All three are repairable from the deposit alone, and none requires an old version of anything: DeclareDesign 1.1.1 still accepts the old assignment and sampling syntax behind `legacy = TRUE`.

1.2 Does the maintained rewrite reproduce the chapter?

Every figure reproduces. What does not is a handful of claims the chapter makes about those figures, most of which belong to the chapter rather than to the code. Of 40 recorded claims, 27 can be checked against the published chapter; 21 hold and 6 do not, four of the latter being sentences or labels the chapter's own data contradict. The 13 remaining are quantities the chapter never states, which is a large share, because this is a methods chapter with no tables and no reported estimates: its figures are drawn from simulated data and its claims are about shape and direction rather than magnitude. The four that are the chapter's own are collected in `coppock_2021_errata.pdf` at the root of this repository; none of them changes a conclusion.

Table 1: Reproduction verdict by component.

Component	Verdict
Figures 17.1, 17.2, 17.6, 17.7, 17.8	Reproduce
Figure 17.3 (blocked, faceted by block)	Figure reproduces; the text describing it transposes the two neighborhoods, and calls the blocks schools
Figure 17.3, panel (b) axis labels	The deposited script abbreviates where the published panel spells out
Figure 17.4 (clustered), estimates	Reproduce
Figure 17.4 (clustered), stated outcome scale	8 of 441 outcomes fall outside the stated 400 to 1600 range
Figure 17.4 (clustered), panel (b) axis label	The deposited script does not produce the published label
Figure 17.5 (covariate adjustment), estimates	Reproduce
Figure 17.5, vertical axis label	Describes a seven-point Likert raw scale; the outcome takes 100 distinct values
Data generator (seven simulated datasets)	48 of 60 columns regenerate identically

All six failures are worth naming, and four of them are the chapter's own. It says the blocked experiment shows "small effects of treatment in neighborhood 1 and large negative effects of treatment in neighborhood 2"; the two neighborhoods are the other way round, in the deposited data, in the deposited generator, and in the published figure. In the next sentence it says Figure 17.3b compares "across schools", of a design with 2 neighborhoods and no schools. It says the clustered outcome is "measured on a 400-1600-point scale"; 8 of the 441 simulated outcomes fall outside it and both panels clip. And the vertical axis of Figure 17.5 calls its raw scale a seven-point Likert, where the outcome takes 100 distinct values across 100 units.

The remaining two are the deposit falling short of the book, in the same place and in the same direction both times: the deposited scripts label two panels less well than the published figures do. `clustered.R` labels both panels of Figure 17.4 as classroom averages, while the published panel (b), which plots students, is labelled "Outcome variable: SAT score". `blocked.R` abbreviates the horizontal axis of Figure 17.3b to "N/hood", while the published panel spells "Neighborhood" out. In both cases the published figure is the better one and running the deposited script does not produce it.

2 Chapter overview

Citation: Coppock, A. (2021). "Visualize as you randomize: Design-based statistical graphs for randomized experiments." In J. N. Druckman and D. P. Green (eds.), *Advances in Experimental Political Science*, 320-336. Cambridge University Press. DOI: 10.1017/9781108777919.022, dated by Crossref to April 2021.

Summary: A methods chapter arguing that a graph of an experiment should encode the design as well as the result. Two design principles do the work: invite visual comparisons across randomly formed groups rather than across groups formed before or after treatment, and show the fitted model and its uncertainty in data space rather than on its own. A third asks that design features like blocking, clustering and differential assignment probabilities be mapped to visual cues. The chapter works through seven designs, each with a panel that follows the advice and a panel that does not: a two-arm trial, a blocked experiment, the same blocked experiment faceted two ways, a cluster-randomized trial, covariate adjustment shown as a residual-on-residual plot, an interaction with a continuous covariate, two-sided noncompliance, and attrition with extreme value bounds. All of the data are simulated, so the chapter reports no estimates and prints no tables. Its empirical content is the figures themselves and the design parameters it states in prose.

3 Original archive reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/VE6VSR. Seventeen files under a single `replication_archive` directory label: eight R scripts, one Stata `.do` file, seven simulated datasets and a README. All seventeen served checksums match the published ones, and the deposit's own README describes the contents accurately, which is not something to take for granted.

Table 2: Original archive reproducibility, checked against R 4.6.0.

Script	Status in a clean R session	Resolution
two_arm_trial.R	Fails: could not find function 'tidy'	library(broom)
blocked.R	Runs	No changes required
clustered.R	Fails: could not find function 'tidy'	library(broom)
covariate_adjustment.R	Fails: could not find function 'lm_lin'	library(estimatr), then library(broom)
noncompliance.R	Fails: could not find function 'tidy'	library(broom)
interaction.R	Runs	No changes required
attrition.R	Fails: could not find function 'tidy'	library(broom)
make_datasets.R	Fails: three separate pre-1.0 DeclareDesign incompatibilities	legacy = TRUE on assignment and sampling; explicit breaks for draw_likert
two_arm_trial.do	Not run: no Stata licence here, and it reads from a data/ path the deposit does not contain	Its arithmetic is reproduced in R and compared against the R figure

Every package the archive names still installs from CRAN: `tidyverse`, `estimatr`, `DeclareDesign`, `MASS`, `margins` and `ggrepel`. Nothing here is unmaintained, and nothing needed substituting for want of a package. Two patterns that commonly break in archives of this vintage do not break here. `do(tidy(...))`, `gather()` and `spread()` emit no warnings at all, being superseded rather than deprecated. And `stat_smooth(method = "lm_robust")` is recognised by ggplot2 4.0.3, because `stat_smooth` resolves a method string with `match.fun()` and `estimatr` is attached by the time the plot is drawn.

Running the archive also leaves seven files behind that the deposit does not contain: six figure PDFs written into whatever directory the scripts were run from, plus an `Rplots.pdf` from the device R opens under `Rscript`. Those scripts were run in a scratch copy for that reason, and `download_original.R` prints a warning if anything but the deposit turns up in `original/`.

3.1 The data generator

`make_datasets.R` is the interesting failure. It declares seven designs in the DeclareDesign syntax of 2019, writes out the seven datasets the analysis scripts read, and stops at the third line of the first design. There are three incompatibilities, and none of them requires an old version:

- `declare_assignment(m = 100)` and its blocked and clustered variants. DeclareDesign 1.0 removed the named-argument form, and its own error message names the replacement. Passing `legacy = TRUE` restores the old behaviour in 1.1.1.
- `draw_likert(0.5 * Z + U)`. `fabricatr` no longer supplies default cut points and now demands `breaks`, or `min`, `max` and `bins`. The cut points the deposit was built with are recoverable from the deposited file, and are the unit-width breaks from -2.5 to 2.5.
- `declare_sampling(prob_unit = pnorm(X), simple = TRUE)`, which fails the same way as the assignment steps and takes the same `legacy = TRUE`.

Repaired, the generator runs to the end, and what it produces is the subject of a later section.

3.2 The Stata file

The deposit's README says its Stata file "produces something similar to Figure 1", which is a claim about output that nobody without a Stata licence can check by running it. The file is nine lines of arithmetic, so `text_stata_equivalent.R` reproduces them in R and compares the result against the intervals the R script draws.

Table 3: The deposited Stata file's intervals against the R script's, on the same data.

Condition	Method	Mean	Lower	Upper
Control	R script: lm_robust HC2, t distribution	0.515	0.466	0.564
Control	Stata .do file: sd / sqrt(N), normal approximation	0.515	0.466	0.564

Treatment	R script: lm_robust HC2, t distribution	0.610	0.513	0.707
Treatment	Stata .do file: sd / sqrt(N), normal approximation	0.610	0.514	0.706

The claim holds. The group means are identical, and the intervals differ only because the Stata file builds a normal-approximation interval from the group standard deviation where the R script builds an HC2 interval on the t distribution. On a 0 to 1 outcome the widths differ by at most 0.0024. What the Stata file cannot do is run as deposited: it reads `data/two_arm_simulated_data.csv` and the deposit has no `data/` directory, so the path has to be repaired before the file will open its own data.

4 Number-by-number comparison

The ground truth is built by `ground_truth/build_ground_truth.R`, which reads every archive and rewrite value out of `maintained/output/` and writes the table. Nothing in it is typed except the values read from the published chapter, and those are only ever comparison targets: no published number is an input to any computation in this repository.

The chapter is unusual in what it gives a reader to check. It has no tables, prints no estimates, and states no standard errors, so most of its checkable content is qualitative: a direction, an ordering, a stated range, an axis label. A claim about a value carries its verdict in the `Match` column; a claim about shape, sign or ordering has no value to compare and carries its verdict in `Holds`. Neither is typed. Both are predicates the build script evaluates against the pipeline output, so that a claim like "the ATE estimate is large and negative" is settled by asking whether the interval excludes zero rather than by an author's judgement about what counts as large.

Table 4: Ground truth: what the chapter states against what the deposited data and scripts produce. Both verdict columns blank means the chapter does not state the quantity.

Location	Claim	Chapter	Archive data	Match	Holds
Figure 17.1	N	500	500	1	
Figure 17.1	N treated	100	100	1	
Figure 17.1	Control group mean		0.515		
Figure 17.1	Treatment group mean		0.610		
Figure 17.1	The deposited Stata file draws something similar to the R figure		intervals differ in width by at most 0.0024		
Figures 17.2 and 17.3	Number of neighborhoods	2	2	1	
Figures 17.2 and 17.3	N in neighborhood 1	50	50	1	
Figures 17.2 and 17.3	N in neighborhood 2	100	100	1	
Figures 17.2 and 17.3	N treated per neighborhood	25	25	1	
Figures 17.2 and 17.3	N total		150		
Figure 17.2	Probability of treatment is higher in the first neighborhood	higher in the first	0.50 against 0.25		1
Figure 17.2	Inverse probability weighted ATE is large and negative	large and negative	-1.284		1
Figure 17.2	Unweighted ATE is close to zero	close to zero	-0.700		1

Table 4: Ground truth: what the chapter states against what the deposited data and scripts produce. Both verdict columns blank means the chapter does not state the quantity. *(continued)*

Location	Claim	Chapter	Archive data	Match	Holds
Figure 17.3	Small effect in neighborhood 1 and a large negative effect in neighborhood 2	small in 1 and large negative in 2	neighborhood 1 = -4.36; neighborhood 2 = 0.25		0
Figure 17.3	Panel b compares across schools	schools	N/hood 1, N/hood 2		0
Figure 17.3	Panel b horizontal axis labels	Neighborhood 1	N/hood 1, N/hood 2	0	
Figure 17.4	Number of classes		30		
Figure 17.4	N students		441		
Figure 17.4	Outcome measured on a 400 to 1600 point scale	400 to 1600	217 to 1720	0	
Figure 17.4	Group means are the same in both panels	the same in both panels	957.9 and 1117.8 in both		1
Figure 17.4	Panel b confidence intervals ignore clustering and are narrower	narrower	widths 59 and 51 against 174 and 107		1
Figure 17.4	Panel b vertical axis label	Outcome variable: SAT score	Outcome variable: Classroom Average SAT score	0	
Figure 17.4	Clustering can be handled by clustered standard errors or by weighting class means	either approach	both give 159.884		1
Figure 17.5	N		100		
Figure 17.5	The vertical scale of both facets is the same but their range is not	the same scale but not the same range	10 units in each facet; 0 to 10 against -5 to 5		1
Figure 17.5	lm_lin is equivalent to interacting the covariate with the treatment indicator	equivalent	1.9796		1
Figure 17.5	The residual on residual slope equals the multiple regression estimate	exactly equal	1.9796		1
Figure 17.5	Vertical axis label describes the raw scale as a seven point Likert	7-point Likert	100 distinct values over 0.6 to 8.7	0	
Figure 17.6	N		1189		
Figure 17.6	Negative effects at low covariate values and positive effects at high ones	negative at low and positive at high	-3.64 at X = -2 and 4.40 at X = 2		1
Figure 17.6	The linear model fits worst at low covariate values	does not fit well, especially at low values	mean residual 10.16 in the lowest bin		1
Figure 17.7	N		600		
Figure 17.7	N per assigned arm		300		
Figure 17.7	Noncompliance is two-sided	two-sided	treatment arm 0.667 and control arm 0.167		1
Figure 17.8	N		200		
Figure 17.8	N missing an outcome		19		
Figure 17.8	Outcome on a seven point Likert scale	seven point Likert	2 to 7	1	
Figure 17.8	Lower bound group means are very similar and the worst case effect is near zero	very similar and close to zero	4.35 and 4.35, a difference of 0.00		1
Figure 17.8	Upper bound effect is close to a full scale point	close to a full-scale point	1.14		1
Figure 17.8	The ATE lies somewhere between zero and one	zero and one	0.00 to 1.14		

Of the 40 recorded claims, 27 state something the chapter can be held to. 21 hold and 6 do not. Every one of the eight published figures carries at least one row.

4.1 The transposed neighborhoods

The chapter's discussion of Figure 17.3 reads:

> In Figure 17.3a, we facet by block. Within each facet, the groups that are compared are formed by random assignment: we see small effects of treatment in neighborhood 1 and large negative effects of treatment in neighborhood 2.

Table 5: Block-level treatment effects in the deposited blocked dataset.

Neighborhood	Residents	Pr(treated)	Effect	SE	p
1	50	0.50	-4.36	0.79	0.000
2	100	0.25	0.25	0.50	0.611

The large negative effect is in neighborhood 1 and the small one is in neighborhood 2. Three independent sources agree: the deposited dataset, the deposited generator, whose potential outcomes are written `-4 * Z * (neighborhood == 1)`, and the published figure itself, whose left facet is labelled "Neighborhood 1" and shows a drop from 10.5 to 6.2. Only the sentence on the facing page has them the other way round. Nothing in the code needs changing and the maintained rewrite reproduces the figure as published; the error is in the prose.

4.2 The blocks that are called schools

The very next sentence reads:

> By contrast, in Figure 17.3b, we facet by randomly assigned group, so we compare across schools and within treatment group.

There are no schools in this design. The blocked example has 2 neighborhoods of 50 and 100 residents, and the horizontal axis of the published Figure 17.3b is labelled "Neighborhood 1" and "Neighborhood 2". The chapter's classroom example is the cluster-randomized experiment two sections later. The word is a slip in a paragraph that otherwise names the neighborhoods correctly four times, but it names a design feature the example does not have, which is why it belongs with the other corrections rather than in a list of typographical points.

The published panel (b) is also where the deposit falls short of the book a second time. `blocked.R` abbreviates the horizontal axis labels of that panel to `N/hood 1` and `N/hood 2` while writing panel (a)'s in full, so running the deposited script does not produce the published labels. Nothing published is wrong here; as with the Figure 17.4 panel (b) label below, the published figure is the better one.

4.3 The 400 to 1600 point scale

The clustered example describes an outcome "measured on a 400-1600-point scale", and both panels of Figure 17.4 set their limits accordingly. The simulated outcomes run from 217 to 1720, so 8 of the 441 students sit outside the stated scale and are clipped out of the figure by `coord_cartesian()`. The design draws a classroom shock at `rnorm(30, 1000, 100)` and a student shock at `rnorm(n, sd = 175)`, which puts about two percent of the distribution outside the range on arithmetic alone. The clipping is the chapter's own choice and the rewrite keeps it; the mismatch is between the data and the sentence that describes them.

4.4 The Likert scale that is not one

The vertical axis of Figure 17.5 reads `Outcome variable (raw scale is 7-point Likert)`. The outcome it plots is not a Likert item and is not seven-point. It takes 100 distinct values across 100 units, running from 0.6 to 8.7, and 5 of them fall outside the 1 to 7 range a seven-point item allows. The generator draws it as `Y ~ 0.5 * Z + 1.0 * X + 0.5 * Z * X + U` with a normal covariate and a normal disturbance, and the unadjusted facet's own axis runs from 0 to 10, which is the shape of a continuous variable rather than an ordinal one. The chapter's only Likert outcome is Figure 17.8's attrition example, which is genuinely seven-point.

`covariate_adjustment.R` writes this label, so the chapter printed what its own code gave it, and the maintained rewrite keeps the label as deposited rather than silently improving it. Nothing plotted in the figure is affected: the correction is to the axis title alone.

4.5 The relabelled panel

`clustered.R` gives both panels the y-axis label `Outcome variable: Classroom Average SAT score`. Panel (b) plots students rather than classes, and the published panel (b) is labelled `Outcome variable: SAT score`. The book restyled every axis label in the chapter to sentence case, so some difference was expected; what makes this one substantive is that the restyling also dropped "classroom average", which is a correction rather than a matter of house style. The published figure is the better one and the deposited code does not produce it.

5 The extraction and the two instruments

A table of claims can only be as complete as the reading behind it, and a table built up figure by figure has no way to notice a sentence nobody thought to check. So the chapter was read a second time as an inventory rather than as an argument: every numeral in the body text, every number spelled out in words, and every quantity a figure asserts without printing it, each recorded with where it appears and what kind of claim it is. That inventory is `ground_truth/published_claims.csv`, 65 rows, hand-reviewed and committed.

Table 6: The extraction. A claim is recomputed when the pipeline can reach the quantity; the rest are verified where they are used, or belong to a cited source rather than to this chapter.

Claim type	Recomputed	Rows
definitional	No	5
definitional	Yes	21
descriptive	Yes	17
structural	No	13
structural	Yes	3
transcribed	No	6

No row is typed `pipeline`, and that is the finding rather than an oversight: the chapter states no estimate anywhere. What it does state is design parameters, response scales, axis labels and claims about the shape of its own results, so a check on this chapter is mostly a check on whether the simulated data behind a figure are what the surrounding sentence says they are. Three of the four errata came out of this pass, and only one of them, the transposed neighborhoods, was reachable by asking what a figure plots.

Six rows are `transcribed`: Fisher's eight cups of tea and the four assigned to milk first, the seventy ways of allocating them, the MIDA framework's four elements, Healy's three dimensions of criticism, and the differencing intuition from Gerber and Green. These belong to the works the chapter cites, cannot drift, and were checked once against those works.

`maintained/in_text_claims.R` is the second instrument. It carries 41 blocks, one for each extraction row the pipeline can reach: the published sentence verbatim in a comment, then code that reads `maintained/output/` and prints the quantity in the chapter's own units, labelled. It recomputes nothing. Estimation happens once, in the figure scripts; only derivation happens twice, and it takes a different route each time. The chapter's two group sizes, for instance, are read out of the deposited dataset by `text_design_parameters.R` and recovered here from the residual degrees of freedom of the two regressions Figure 17.1 plots. The confidence level the chapter calls 95 per cent is recovered from the interval half-widths rather than asserted. The imputation range behind the extreme value bounds is recovered from the distance between the two bounds and the number of missing outcomes, without knowing how those missing outcomes split across arms.

`build_ground_truth.R` ends by running that file as a program, not by reading it as text, since a block that errors or prints nothing would satisfy any textual check. It captures what the file prints and requires three things: that the number of claims printed equal the number of extraction rows needing a block, and that the two sets of identifiers be equal in both directions; that the chapter's own values, transcribed independently into the extraction and into the ground truth, agree string for string; and that wherever both instruments arrive at a bare number, they agree at the precision the extraction records for that claim. The claims file is sourced into its own environment, because the two files necessarily read the same outputs and name the same things, and a bare `source()` would replace the build's objects with the claims file's before any of those assertions ran.

Of the failing claims, four fail in a way that is the chapter's own rather than the deposit's or the environment's. Those are set out in `coppock_2021_errata.pdf` at the root of this repository: the published sentence, the corrected sentence with the changed token in bold, and the evidence. Every number in that document is computed from `maintained/output/` when it is rendered, including the numbers that were not wrong. None of the four changes a conclusion of the chapter.

6 Maintained rewrite

The rewrite lives in `maintained/`: seven figure scripts, four scripts for quantities that belong to no single figure, a port of the data generator, and a shared `helpers.R`. It is a translation, not a reanalysis: every estimator and every plotted quantity is the one the chapter used.

Table 7: Maintained rewrite scripts.

Script	Produces
figure_1_two_arm_trial.R	Figure 17.1, both panels
figure_2_blocked_experiment.R	Figures 17.2 and 17.3, four panels from one dataset
figure_4_clustered_experiment.R	Figure 17.4, both panels
figure_5_covariate_adjustment.R	Figure 17.5
figure_6_interaction_continuous.R	Figure 17.6, both panels
figure_7_noncompliance.R	Figure 17.7, both panels
figure_8_attrition.R	Figure 17.8
text_design_parameters.R	The design facts the chapter states in prose
text_descriptive_claims.R	The three response scales the chapter states a bound for
in_text_claims.R	The claim-by-claim audit trail, and the coverage gate's input
text_stata_equivalent.R	The deposited Stata file's estimator, in R, against the R script's
text_archive_agreement.R	Every figure's estimates re-derived from the deposit, against the rewrite's
make_datasets.R	The seven simulated datasets, regenerated under the current DeclareDesign API

Every figure script also writes the estimates it plots to a CSV, along with the axis labels and panel ranges the chapter makes claims about. The chapter prints no numbers, so an output folder of images alone would make the reproduction diff a formality: a PDF differs on every run because it records the time it was written. Writing the plotted quantities out is what makes `git diff` on `maintained/output/` a check rather than a ritual.

6.1 Deprecated patterns replaced

Table 8: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘library()‘ in each analysis script	‘source(here::here(”maintained”, ”helpers.R”))‘
relative paths to the working directory	‘here::here()‘
‘do(tidy(lm_robust(..., data = .)))‘	‘reframe(tidy(lm_robust(..., data = pick(everything()))))‘
‘gather()‘ / ‘spread()‘	‘pivot_longer()‘ / ‘pivot_wider()‘
‘geom_errorbar(width = 0)‘	‘geom_linerange()‘
‘data.frame()‘ for small hand-built frames	‘tibble()‘
‘summary() |> as.data.frame()‘	‘as_tibble()‘
‘declare_population()‘ / ‘declare_assignment(m =)‘	‘declare_model()‘ / ‘declare_assignment(Z = complete_ra(...))‘
figures written into the archive directory	‘ggsave()‘ to ‘maintained/output/‘
magrittr pipe	native pipe

One substantive correction was made. The archive’s `interaction.R` writes `geom_errorbar(aes(ymax = lower, ymin = upper), width = 0)` with the bounds transposed. Because `width = 0` draws a bare segment with no caps, the transposition is invisible in the output, and the rewrite writes them the right way round. No published value changes.

6.2 Three things left alone

The `coord_cartesian()` limits that clip eight clustered observations are the chapter’s, and they are kept. Widening them would produce a figure the book does not contain.

`figure_5_covariate_adjustment.R` residualises with `lm()` and then plots, exactly as the deposit does, rather than reading the coefficients off `lm_lin()`. The archive prints three estimators side by side under the comment “standard errors v. slightly off; point estimates OK”, and the rewrite writes those three out instead of printing them into a console that no longer exists. Both halves of the comment hold: the point estimates agree to fifteen digits at 1.9796, which is what the chapter says they should do, and the standard errors are 0.09486 against 0.09398.

The unseeded `position_jitter()` calls are seeded at 42. That moves the point clouds slightly relative to the published figures and is the only respect in which any panel differs from the book beyond the axis label discussed above. Seeding a jitter changes an irreproducible figure into a reproducible one; it does not change what the figure shows.

7 The rewrite against the archive

The chapter states no estimates, so a claim that the rewrite reproduces the archive cannot rest on comparing published numbers. `text_archive_agreement.R` settles it directly: it re-derives every estimate behind the eight figures from the deposited data, using the model specifications the deposited scripts use, and compares the result against what the figure scripts wrote. The reshaping and labelling deliberately differ from the deposit, because that is where a faithful-looking port goes wrong; the model calls do not.

Table 9: The rewrite's plotted estimates against the same quantities re-derived from the deposit.

Figure	Quantity	Values	Largest absolute difference
Figure 17.1	Group means and intervals	8	5.55e-17
Figure 17.2	Weighted and unweighted group means	16	5.55e-17
Figure 17.3	Group means within each facet	32	0.00e+00
Figure 17.4	Clustered and unclustered group means	16	2.27e-13
Figure 17.5	Three covariate adjustment estimators	12	2.22e-16
Figure 17.6	CATE across the covariate grid	51	4.44e-16
Figure 17.7	Group means by assigned group	16	2.78e-17
Figure 17.8	Bounded group means	16	4.44e-16

All 167 values agree to within 2.3e-13, which is floating point noise. The deposited scripts' own `do(tidy(...))` idiom returns the same numbers. The ground truth builder asserts this agreement and stops if it ever fails, which is why its archive and rewrite columns carry the same value.

8 The data generator under the current API

`maintained/make_datasets.R` declares the same seven designs using `declare_model()` and the current assignment functions, seeds at 343 as the archive does, and compares every regenerated column against the deposited file.

Table 10: The current-API port against the deposited simulated data.

Dataset	Columns	Identical	Columns that differ
attrition	10	7	Z, R, Y
blocked	9	9	none
clustered	11	8	Z, Y, condition
covariate	9	6	Z, Y, condition
interaction	10	7	Z, Y, condition
noncompliance	3	3	none
two_arm	8	8	none

48 of the 60 regenerated columns come back identical to the deposit. Every fabricated covariate and every potential outcome reproduces, in all seven datasets. What differs is the realised assignment vector in four of them, and the columns revealed through it. The pattern is specific: `complete_ra(N, m = 100)` and `block_ra(blocks, m = 25)` return the assignments the deposit records, while `complete_ra(N)` with no `m` and `cluster_ra(clusters)` do not. The drift is in `randomizr`, not in R's sampler. Running the repaired generator under `RNGkind(sample.kind = "Rounding")`, which restores the pre-3.6 `sample()`, makes matters worse rather than better: the blocked dataset stops reproducing too. Since the analysis scripts read the deposited files rather than these, no published figure depends on the difference.

Two details of the port are worth recording because they are the reason it reproduces anything at all.

The reveal steps use `declare_reveal()` rather than `declare_measurement()` with `reveal_outcomes()`. The two are equivalent here, but the second draws two extra random numbers per design, which moves every design after the first off the archive's random number stream and makes the comparison above uniformly negative for reasons that have nothing to do with the deposit.

The generator's blocked section prints two regressions before moving on, and the inverse-probability-weighted one draws two random numbers on its way through `estimatr`. Those draws sit in the stream between the blocked design and every design declared after it, so the deposited data depend on a diagnostic call that computes nothing the file needs. The port keeps the pair and writes their output out rather than printing it. This is a real property of the deposit worth stating plainly: five of its seven datasets cannot be regenerated by any script that tidies away a dead `print()`.

9 Figure verification

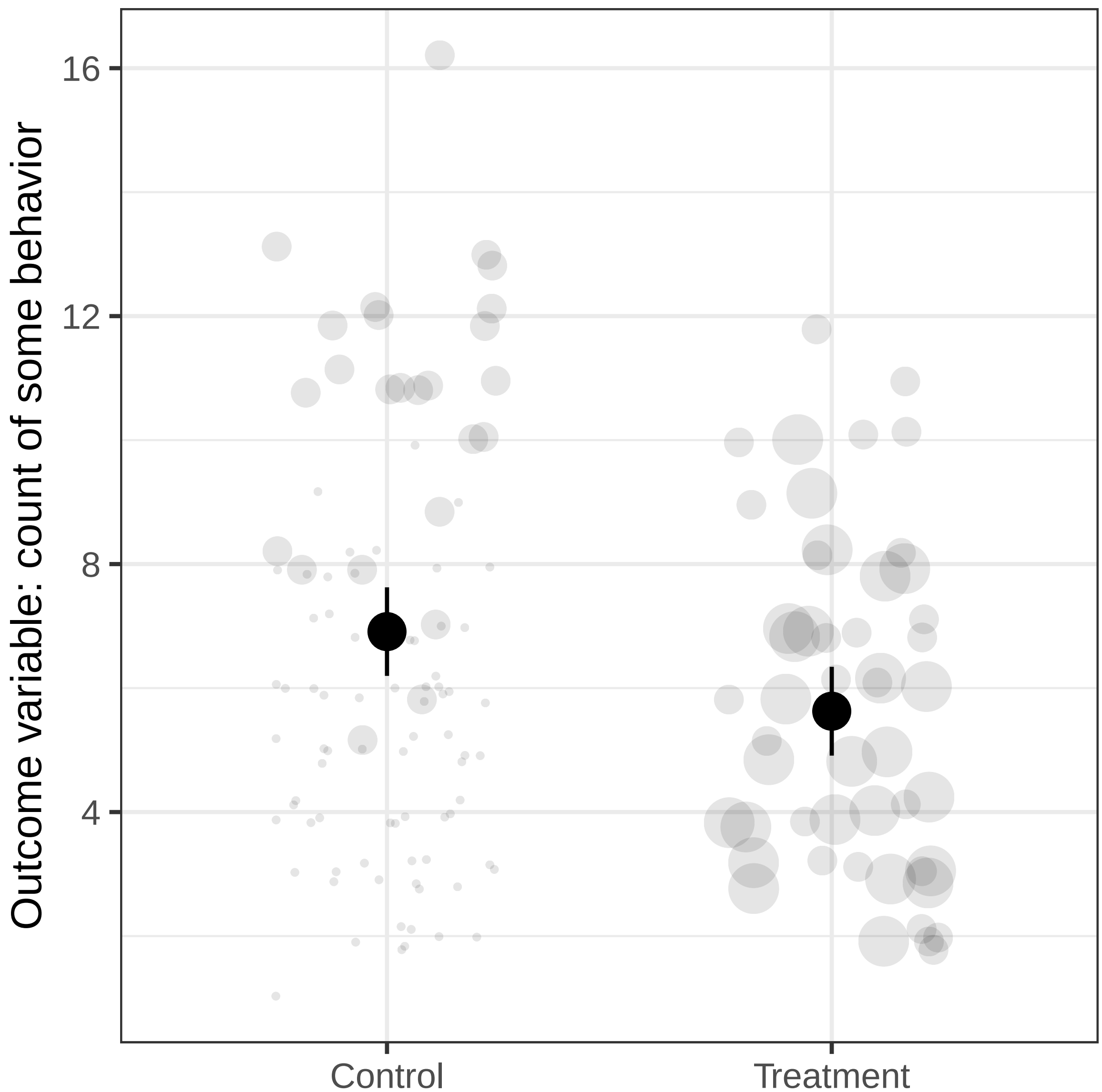

Figure 1: Figures 17.2 and 17.3 as reproduced by the maintained rewrite: the blocked experiment weighted and unweighted, then faceted by block and by condition.

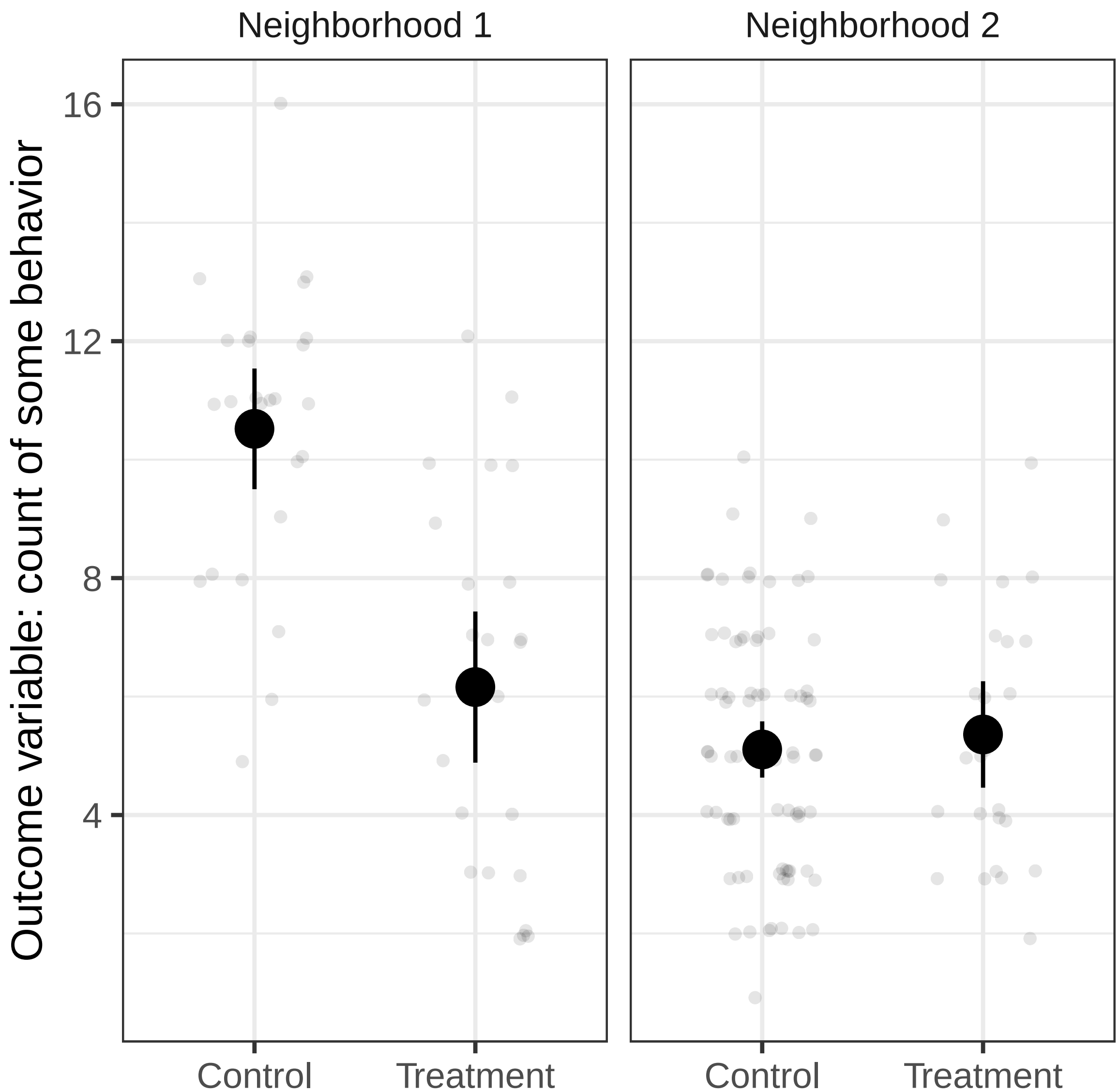

Figure 2: Figures 17.2 and 17.3 as reproduced by the maintained rewrite: the blocked experiment weighted and unweighted, then faceted by block and by condition.

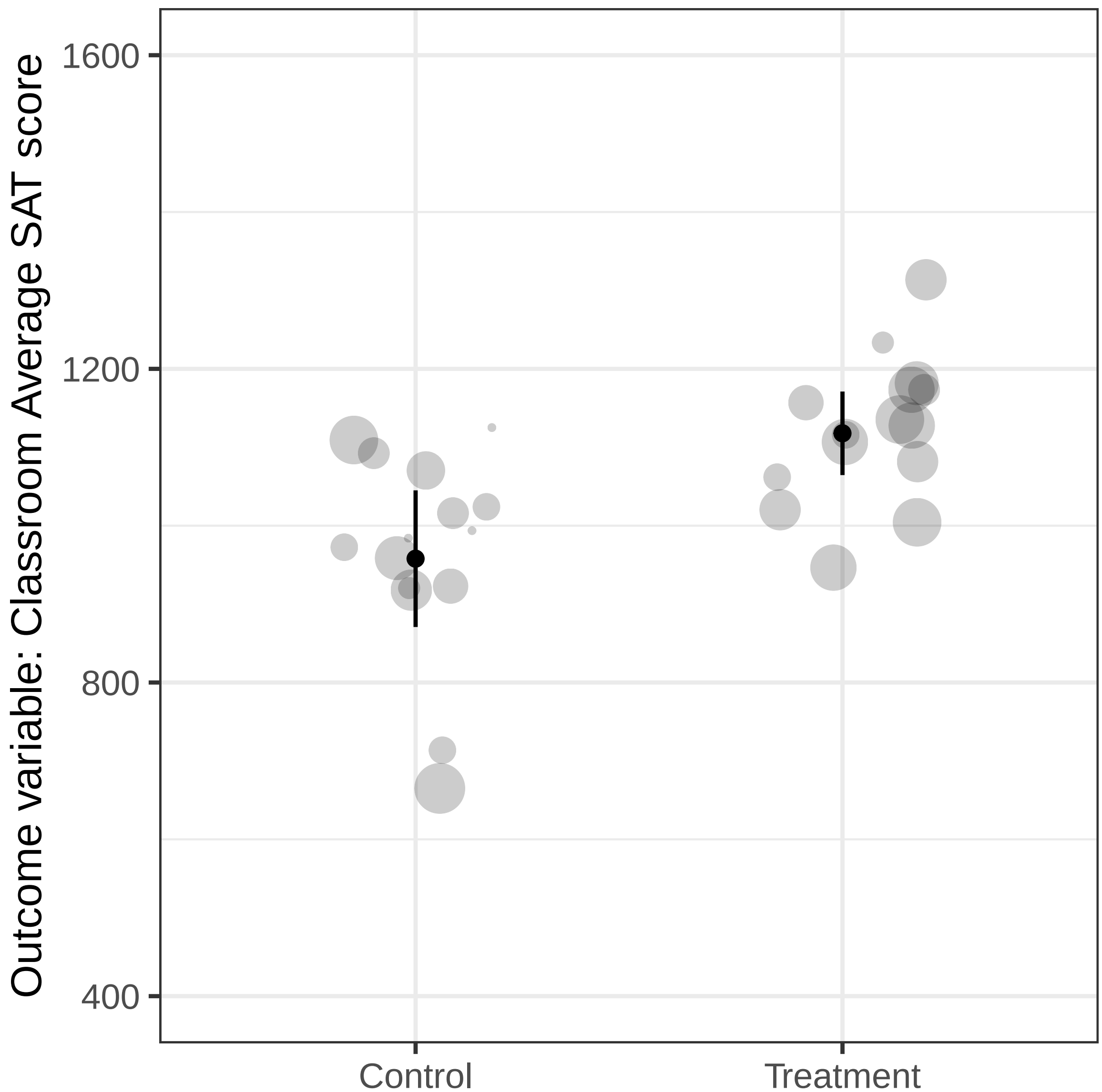

Figure 3: Figure 17.4 as reproduced by the maintained rewrite: cluster means with cluster-robust intervals, then students with intervals that ignore the clustering. Both carry the deposited script's axis label; the published panel (b) does not.

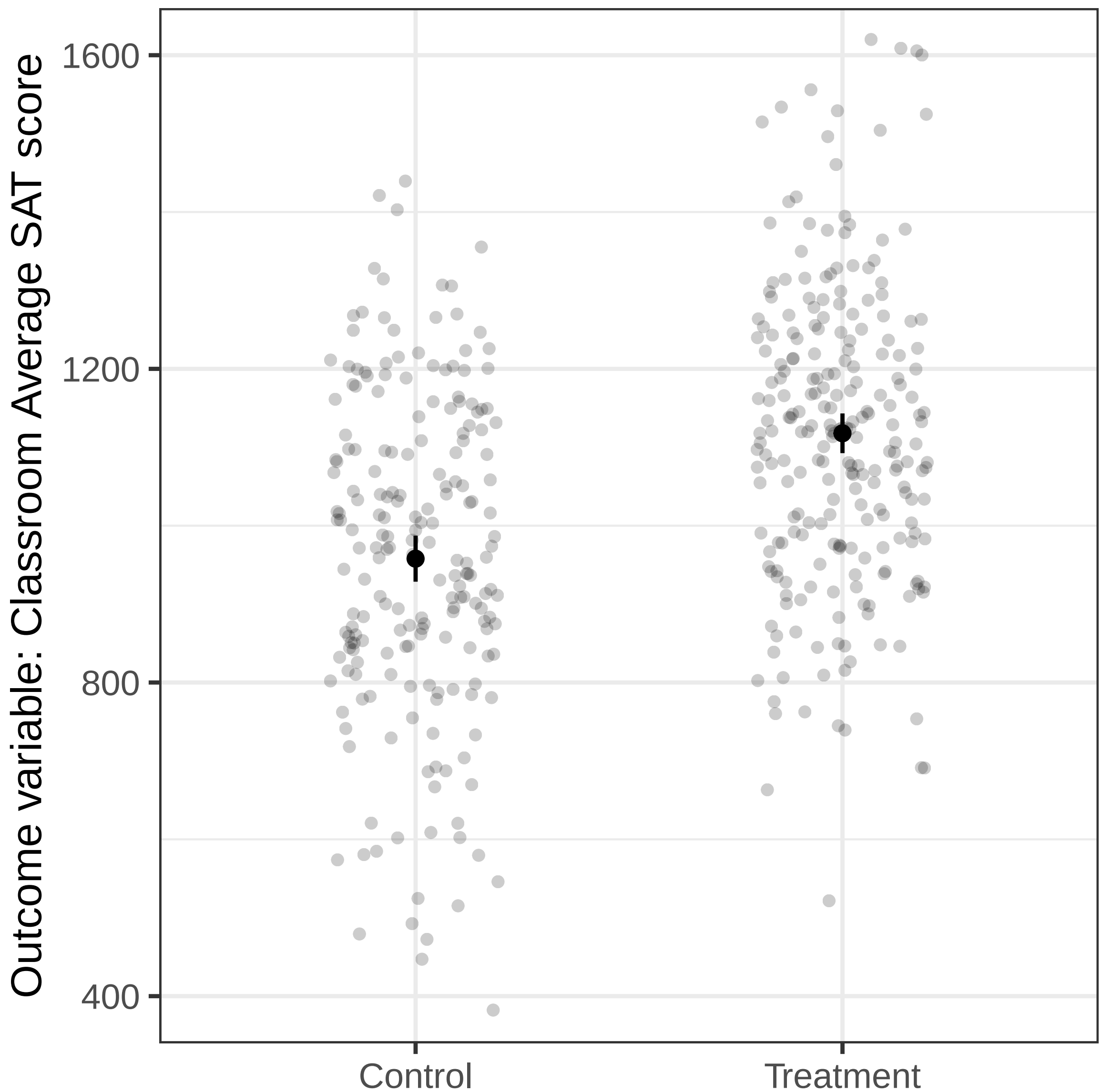

Figure 4: Figure 17.4 as reproduced by the maintained rewrite: cluster means with cluster-robust intervals, then students with intervals that ignore the clustering. Both carry the deposited script's axis label; the published panel (b) does not.

All fourteen panels reproduce the published figures. Beyond the seeded jitter, the two axis labels the book improved on, and the Figure 17.5 axis label that neither the book nor the deposit got right, no panel differs from its published counterpart.

Table 11: Design parameters read off the deposited datasets by the in-text script.

Figure	Quantity	Value in the deposited data
Figure 17.1	N	500
Figure 17.1	N treated	100
Figure 17.1	Number of distinct outcome values	2
Figure 17.1	Outcome range	0 to 1
Figures 17.2 and 17.3	N	150
Figures 17.2 and 17.3	Number of neighborhoods	2
Figures 17.2 and 17.3	N in neighborhood 1	50
Figures 17.2 and 17.3	N in neighborhood 2	100
Figures 17.2 and 17.3	N treated per neighborhood	25
Figures 17.2 and 17.3	Outcome range	1 to 16
Figures 17.2 and 17.3	Number of non-integer outcomes	0
Figure 17.4	N students	441
Figure 17.4	Number of classes	30
Figure 17.4	Smallest class	10
Figure 17.4	Largest class	20
Figure 17.4	Outcome range	217 to 1720
Figure 17.5	N	100
Figure 17.5	Number of distinct outcome values	100
Figure 17.5	Outcome range	0.6 to 8.7
Figure 17.6	N	1189
Figure 17.7	N	600
Figure 17.7	N assigned to treatment	300
Figure 17.7	N assigned to control	300
Figure 17.8	N	200
Figure 17.8	N missing an outcome	19
Figure 17.8	Outcome range	2 to 7

10 Maintained rewrite verification

Table 12: Maintained rewrite verification: what the chapter states against what the rewrite produces.

Location	Claim	Chapter	Rewrite	Match	Locus
Figure 17.1	N	500	500	1	
Figure 17.1	N treated	100	100	1	
Figure 17.1	Control group mean		0.515		
Figure 17.1	Treatment group mean		0.610		
Figure 17.1	The deposited Stata file draws something similar to the R figure		intervals differ in width by at most 0.0024		
Figures 17.2 and 17.3	Number of neighborhoods	2	2	1	
Figures 17.2 and 17.3	N in neighborhood 1	50	50	1	
Figures 17.2 and 17.3	N in neighborhood 2	100	100	1	
Figures 17.2 and 17.3	N treated per neighborhood	25	25	1	
Figures 17.2 and 17.3	N total		150		
Figure 17.2	Probability of treatment is higher in the first neighborhood	higher in the first	0.50 against 0.25		

Table 12: Maintained rewrite verification: what the chapter states against what the rewrite produces. *(continued)*

Location	Claim	Chapter	Rewrite	Match	Locus
Figure 17.2	Inverse probability weighted ATE is large and negative	large and negative	-1.284		
Figure 17.2	Unweighted ATE is close to zero	close to zero	-0.700		
Figure 17.3	Small effect in neighborhood 1 and a large negative effect in neighborhood 2	small in 1 and large negative in 2	neighborhood 1 = -4.36; neighborhood 2 = 0.25		paper_internal
Figure 17.3	Panel b compares across schools	schools	N/hood 1, N/hood 2		paper_internal
Figure 17.3	Panel b horizontal axis labels	Neighborhood 1	N/hood 1, N/hood 2	0	archive
Figure 17.4	Number of classes		30		
Figure 17.4	N students		441		
Figure 17.4	Outcome measured on a 400 to 1600 point scale	400 to 1600	217 to 1720	0	paper_internal
Figure 17.4	Group means are the same in both panels	the same in both panels	957.9 and 1117.8 in both		
Figure 17.4	Panel b confidence intervals ignore clustering and are narrower	narrower	widths 59 and 51 against 174 and 107		
Figure 17.4	Panel b vertical axis label	Outcome variable: SAT score	Outcome variable: Classroom Average SAT score	0	archive
Figure 17.4	Clustering can be handled by clustered standard errors or by weighting class means	either approach	both give 159.884		
Figure 17.5	N		100		
Figure 17.5	The vertical scale of both facets is the same but their range is not	the same scale but not the same range	10 units in each facet; 0 to 10 against -5 to 5		
Figure 17.5	lm_lin is equivalent to interacting the covariate with the treatment indicator	equivalent	1.9796		
Figure 17.5	The residual on residual slope equals the multiple regression estimate	exactly equal	1.9796		
Figure 17.5	Vertical axis label describes the raw scale as a seven point Likert	7-point Likert	100 distinct values over 0.6 to 8.7	0	paper_internal
Figure 17.6	N		1189		
Figure 17.6	Negative effects at low covariate values and positive effects at high ones	negative at low and positive at high	-3.64 at X = -2 and 4.40 at X = 2		
Figure 17.6	The linear model fits worst at low covariate values	does not fit well, especially at low values	mean residual 10.16 in the lowest bin		
Figure 17.7	N		600		
Figure 17.7	N per assigned arm		300		
Figure 17.7	Noncompliance is two-sided	two-sided	treatment arm 0.667 and control arm 0.167		
Figure 17.8	N		200		
Figure 17.8	N missing an outcome		19		
Figure 17.8	Outcome on a seven point Likert scale	seven point Likert	2 to 7	1	
Figure 17.8	Lower bound group means are very similar and the worst case effect is near zero	very similar and close to zero	4.35 and 4.35, a difference of 0.00		
Figure 17.8	Upper bound effect is close to a full scale point	close to a full-scale point	1.14		
Figure 17.8	The ATE lies somewhere between zero and one	zero and one	0.00 to 1.14		

The rewrite and the deposited data give the same answer to every claim, which they should: both read the same seven deposited CSV files, and the previous section measures how closely. Two of the three failures sit in the chapter's prose and no rewrite of the code can repair them; the third sits in the deposited script, where the published figure carries a label the code does not produce.

11 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-03

Table 14: Package versions used for the run behind this report.

Package	Version
tidyverse	2.0.0
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
estimatr	1.0.6
broom	1.0.13
margins	0.3.28
DeclareDesign	1.1.1
randomizr	1.0.1
fabricatr	1.0.2
here	1.0.2